Learning-Based Minimally-Sensed Fault-Tolerant Adaptive Flight Control

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***Abstract*—Many multirotor aircraft use redundant configurations to maintain control in the event of an actuator failure. Due to the redundancy of the system, fault isolation is inherently difficult and further compounded by complex interacting aerodynamics of the propellers, wings, and body. This paper presents a novel sparse failure identification and control correction method that does not require direct fault sensing, and instead utilizes only the vehicle's dynamic response. The method couples an ℓ_1-regularized representation of the failure with a deep neural network to effectively isolate faults and improve tracking control in highly dynamic environments with unmodeled aerodynamic effects and unknown actuator failures. The method also includes a control re-allocation scheme which corrects for the identified faults while maximizing control authority and maintaining nominal performance characteristics. Experimental results demonstrate the method's ability to maintain control of a multirotor aircraft by isolating motor failures and reallocating control, improving position tracking by 48 % over the baseline. This paper contributes to the development of robust fault detection and control strategies for over-actuated aircraft.**

***Index Terms*—Failure Detection and Recovery, Machine Learning for Robot Control, Robust/Adaptive Control**

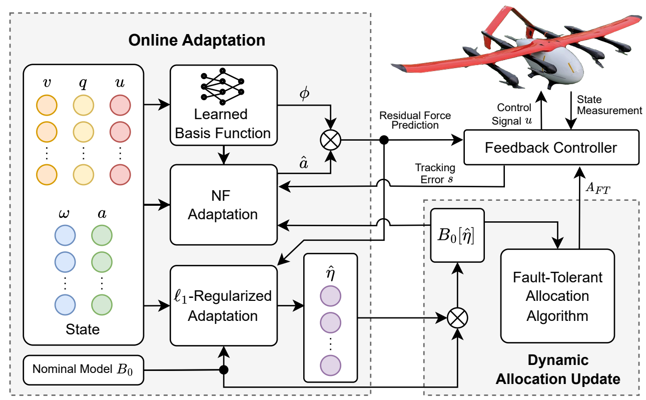

Fig. 1. NFFT combines sparse actuator failure identification with adaptive control using residual force estimation. Failures are identified without direct sensing of the actuator states through an ℓ_1-regularized adaptive update policy, allowing the system to compensate for faults by dynamically updating the control allocation matrix.

I. Introduction

Aerial robots, such as uninhabited aerial vehicles (UAVs) and passenger-fairing electric Vertical Take-off and Landing (eVTOL) aircraft, are becoming increasingly commonplace, emphasizing the need for robust Fault Detection, Isolation, and Recovery (FDIR) methods. Operating these aircraft poses significant challenges and risks, particularly when navigating above densely populated areas where a major malfunction could jeopardize lives and assets. Thus, there is a great interest in improving the survivability of these aircraft to events such as motor and actuator failures.

One method to enhance aircraft resilience to actuator failure is to employ highly redundant control actuation schemes. Although this redundancy intends to secure sufficient control authority for post-failure recovery, it also increases control system complexity and complicates failure identification as many actuators have similar effects on the system. The common approach for fault diagnosis involves fault tree analysis and sensor-based monitoring of predictable failure points. However, as system complexity grows, so does the likelihood of unforeseen failures. Challenges arise from issues like loose connections, calibration errors, software glitches, and malicious interference, which are difficult to address in real-time for autonomous systems. Sensors may not be able to detect complex interactions or dependencies between multiple subsystems or components, which could lead to cascading failures or unforeseen emergent behavior in the system as a whole. This presents a hurdle in designing fault-tolerant control (FTC) systems capable of managing potential failures.

To address these challenges, this paper presents a method for robust, data-driven FTC called Neural-Fly for Fault Tolerance (NFFT) (Fig. 1). The major contributions of NFFT are 1) a minimally-sensed sparse fault identification algorithm that does not require direct actuator sensing and is fully integrated with the Neural-Fly adaptive controller [1] allowing for rapid correction of failure-dependent neural network model of disturbance; 2) a novel control allocation scheme that leverages identified faults to redesign the control allocation matrix in real-time; and 3) validation through real-world flight tests to provide compelling examples of the benefits of this work. Although this study focuses specifically on actuator failures as an exemplar, NFFT can be applied to a wider range of typically non-sensed failure modes such as blade damage or motor movement in flight.

Related work. Early studies on fault diagnosis [2], [3] use Interacting Multiple Models, propagating the probabilities of

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various failure scenarios and applying a probability weighted fusion of the ideal estimated state or control command for each identified failure. Recent surveys [4], [5] show that newer methods often identify the most likely fault and perform discrete switches between nominal and fault-compensating control schemes [6]–[8]. [9] builds on adaptive control tools [10], though requires Persistency of Excitation (PE) (see [11], [12]).

Recent data-driven approaches for fault diagnosis [13], [14] require tailored sensor data, such as electrical current, motor speeds, or acoustic noise measurements. [15] presents FTC scheme utilizing supervisory reinforcement learning while [16] employs adaptive NN-based FTC with automatically regulated neural network weights. The efficacy of both methods have been shown in simulation, but lack experimental validation.

Most prior work focuses on recovery without considering the preceding steps for fault diagnosis. For known faults that leave the aircraft under-actuated, [17]–[19] employ strategies that allow the aircraft to rotate about a fixed axis in order to maintain position. For over-actuated systems, control allocation algorithms highlighted by [20]–[26] prove particularly valuable. Not only do these algorithms aid in designing robust vehicle configurations [21], [22], but they also enable real-time response to identified faults [20]–[22], [25], [26]. Similarly, [27] presents methods for analyzing vehicle performance limits under specific failures.

Robust control and adaptive control algorithms have been used for fault diagnosis and FTC. For benign failures such as partial propeller damage, icing, battery voltage drop and blade pitch locking, methods such as [1], [28]–[32] have shown to be useful. These strategies deliver good nominal condition performances but lose effectiveness with substantial plant changes, such as complete motor failures.

Finally, composite adaptation of combining online parameter estimation and tracking-error adaptive control was introduced first in [10], [33]. Incorporation of online least squares necessitates comparison to more fundamental ideas like sparse solutions in the least squares problems for fault diagnosis. [34] shows equivalence of Lasso shrinkage, coined by [35], and ℓ_1-regularization, which both find sparse-solutions.

Despite the existing methods for FDIR and FTC, a substantial gap persists in dealing with unforeseen failures not accounted for during the design phase of fault mitigation systems. This paper bridges the gap with NFFT, which utilizes only the aircraft's response to control commands to isolate failures, eliminating the need for direct sensing of a predetermined, enumerated set of potential failure modes. Additionally, NFFT reconstitutes the original performance of the system as closely as possible for both partial and complete motor failures. Finally, NFFT extends [1] to rapidly adapt a deep neural network that learns model mismatch due to complex aerodynamics such as propeller interactions and time-varying wind, enhancing both tracking control performance and failure isolation. To our knowledge, NFFT is the first adaptive controller that uses ℓ_1 regularization, overcoming the complexity of the closed-loop control analysis from the non-smooth behavior of the norm.

II. Nominal System Dynamics and Control Laws

Consider a multirotor with dynamics of the form

$$\dot{x} = f(x) + B_0 u + g(x,u,t) + d(t) \tag{1}$$

where $x \in \mathbb{R}^n$ is the n-dimensional state vector, $f(x) \in \mathbb{R}^n$ is the estimated nominal dynamics model, $u \in [0,1]^m$ is the normalized m-dimensional actuator control input, B_0 is the $n \times m$ nominal control actuation matrix, determined from the propeller aerodynamic characteristics and the vehicle configuration, and $g(x,u,t) \in \mathbb{R}^n$ is the unknown residual dynamics caused by unknown actuator faults as well as by unknown external disturbances. While $g(x,u,t)$ is estimated and updated using the NFFT method (see Sec. III), $d(t) \in \mathbb{R}^n$ is any remaining modelling error or disturbance.

The measurement y of $g(x,u,t)$ with noise δ is given as

$$y = g(x,u,t) + \delta. \tag{2}$$

For small $g(x,u,t)$ and $d(t)$ in the nominal operating condition, the system is exponentially stabilized by

$$u = {B_0}^{\text{-}R}(-K\tilde{x} + \dot{x}_d - f(x)), \tag{3}$$

where $K > 0$, x_d is the desired state, $\tilde{x}$ is $x - x_d$, and ${B_0}^{\text{-}R}$ is any right inverse of B_0 such that $B_0{B_0}^{\text{-}R} = I$.

III. Neural-Fly Adaptation with Fault Tolerance

Under actuator faults, $g(x,u,t)$ defined in (1) becomes

$$\begin{aligned} g(x,u,t) &= B_0[\eta]u - B_0 u + r(x,u,t) \\ &= B_0([\eta] - I)u + r(x,u,t) \end{aligned} \tag{4}$$

where $\eta(t) = (\eta_1, \cdots, \eta_m)^\top$, $\eta_i(t) \in [0,1]$ is the effectiveness of the ith actuator at time t with $\eta_i = 0$ corresponds a complete failure, $[\eta] = \operatorname{diag}(\eta(t))$, and $r(x,u,t)$ denotes remaining residual forces not captured by η. The residual dynamics, $g(x,u,t)$, are approximated by

$$\hat{g}_{\text{NFFT}}(x,u,t) = B_0([\hat{\eta}] - I)u + \phi(x,u)\hat{a}(t) \tag{5}$$

where $\hat{\eta}$ is the estimated actuator effectiveness, $[\hat{\eta}] = \operatorname{diag}(\hat{\eta})$, and $\phi(x,u)\hat{a}(t)$ is Neural-Fly's learned residual model of the time-varying aerodynamic forces [1]. Importantly, $\hat{g}_{\text{NFFT}}$ is the sum of Neural-Fly's residual dynamics [1] and $B_0([\hat{\eta}] - I)u$, the estimate of model mismatch due to actuator faults. (5) can be viewed as a first order expansion of the neural network dynamics, which addresses inherent limitations to residual-learning control introduced in [36]. This yields the control law u for (1) with the form

$$u = u_{\text{NFFT}} = (B_0[\hat{\eta}])^{\text{-}R}(-K\tilde{x} + \dot{x}_d - f(x) - \phi\hat{a}) \tag{6}$$

including both the residual model and actuator effectiveness.

A. Sparse Failure Identification

First, consider the simplified model of g and control law u for (1) when residual and disturbance forces are small:

$$\begin{aligned} &\hat{g}_{\text{FT}}(x,u) = B_0([\hat{\eta}] - I)u, \\ &u = u_{\text{FT}} = (B_0[\hat{\eta}])^{\text{-}R}\left(-K\tilde{x} + \dot{x}_d - f(x)\right) \end{aligned} \tag{7}$$

where $\hat{g}_{\text{FT}} \approx g(x,u,t) - r(x,u,t)$ denotes the estimated model mismatch due to actuator faults only. In Sec. III-B, $r(x,u,t)$ is accounted for by $\hat{g}_{\text{NF}}$ (26), as proposed in [1], which utilizes the learned residual dynamics model $\phi(x,u)\hat{a}$, where $\phi(x,u)$ is the meta-trained time-invariant representation of the residual dynamics and $\hat{a}$ is the estimated linear weights that is updated through online composite adaptation. In Sec. III-C, $\hat{g}_{\text{FT}}$ and $\hat{g}_{\text{NF}}$ are combined to yield $\hat{g}_{\text{NFFT}}$ defined in (5).

This scheme (7) estimates $\hat{\eta}$ online without sensing the actuator states directly, allowing computation of the optimal control allocation $A = (B_0[\hat{\eta}])^{\text{-}R}$ to reconstitute nominal performance characteristics in the presence of actuator failure.

In an over-actuated system, a failure cannot be uniquely identified purely from the dynamic response of the system. Hence, an ℓ_1-regularized adaptive update policy [35] is employed for such estimation to encourage sparse solutions [34]. Thus, there is no hard constraint on the number of non-zero parameters or iteration through many non-zero parameter combinations like many prior FTC studies [6]–[8].

1) Continuous Update Law: Consider the following least squares cost function for continuous time t

$$J(\hat{\eta}) = \int_0^t \mathrm{e}^{-\omega_f(t-r)} \left\| y - \hat{g}_{\text{FT}} \right\|^2 \mathrm{d}r + 2\gamma \|\hat{\eta} - 1\|_1 \tag{8}$$

where ω_f is the exponential forgetting rate [10] and γ is the regularization factor. Letting $\bar{\eta} = \hat{\eta} - 1$, $U = \operatorname{diag}(u)$, and $\hat{g}_{\text{FT}} = B_0 U \bar{\eta}$, (8) is rearranged to take the form

$$J(\bar{\eta}) = \int_0^t \mathrm{e}^{-\omega_f(t-r)} \left\| y - B_0 U(r)\bar{\eta} \right\|^2 \mathrm{d}r + 2\gamma \|\bar{\eta}\|_1 \tag{9}$$

Note $\|\bar{\eta}\|_1$ is approximated such that for i actuators and a small positive scalar ϵ, $\|\bar{\eta}\|_1 \approx \sum_i \sqrt{\bar{\eta}_i^2 + \epsilon}$ and $\lim_{\epsilon \to 0} \sum_i \sqrt{\bar{\eta}_i^2 + \epsilon} = \|\bar{\eta}\|_1$.

Then the cost function (9) is approximated $J(\bar{\eta}, \epsilon)$ such that $\lim_{\epsilon \to 0} J(\bar{\eta}, \epsilon) = J(\bar{\eta})$, where $J(\bar{\eta}, \epsilon)$ is given by

$$J(\bar{\eta}, \epsilon) = \int_0^t \mathrm{e}^{-\omega_f(t-r)} \left(y^\top y - 2y^\top B_0 U \bar{\eta} + \bar{\eta}^\top U B_0^\top B_0 U \bar{\eta} \right) \mathrm{d}r + 2\gamma \sum_i \sqrt{\bar{\eta}_i^2 + \epsilon} \tag{10}$$

Since this cost function is convex in $\bar{\eta}$, the minimum value is obtained when $\frac{\partial J}{\partial \bar{\eta}} = 0$, leading to

$$\frac{\partial J}{\partial \bar{\eta}} = \int_0^t \mathrm{e}^{-\omega_f(t-r)} \left(-2UB_0^\top y + 2UB_0^\top B_0 U \bar{\eta} \right) \mathrm{d}r \tag{11}$$

$$\bar{\eta} = P_\eta \int_0^t \mathrm{e}^{-\omega_f(t-r)} U(r) B_0^\top y \mathrm{d}r \quad \text{where } P_\eta = \tag{12}$$

$$\left(\gamma \cdot \operatorname{diag}\left[\tfrac{1}{\sqrt{\bar{\eta}_1^2 + \epsilon}}, \tfrac{1}{\sqrt{\bar{\eta}_2^2 + \epsilon}}, \cdots \right] + \int_0^t \mathrm{e}^{-\omega_f(t-r)} U(r) B_0^\top B_0 U(r) \mathrm{d}r \right)^{-1}$$

If $\bar{\eta}_i = 0$ for some i (typically for ℓ_1 norm minimization), then $\frac{1}{\sqrt{\bar{\eta}_i^2 + \epsilon}} = \frac{1}{\sqrt{\epsilon}}$. Otherwise, as $\lim_{\epsilon \to 0} \frac{1}{\sqrt{\bar{\eta}_i^2 + \epsilon}} = \frac{1}{|\bar{\eta}_i|}$.

By using $\dot{P}_\eta = -P_\eta \frac{\mathrm{d}\left(P_\eta^{-1}\right)}{\mathrm{d}t} P_\eta$,

$$\dot{P}_\eta = \omega_f P_\eta - P_\eta \left(\gamma \mathcal{G}(\bar{\eta}, \epsilon) + U(t)^\top B_0^\top B_0 U(t) \right) P_\eta \tag{13}$$

where $\mathcal{G}(\bar{\eta}, \epsilon) = \operatorname{diag}\left[\frac{\bar{\eta}_1 + \omega_f \bar{\eta}_1^2 + \omega_f \epsilon}{\left(\bar{\eta}_1^2 + \epsilon\right)^{3/2}}, \frac{\bar{\eta}_2 + \omega_f \bar{\eta}_2^2 + \omega_f \epsilon}{\left(\bar{\eta}_2^2 + \epsilon\right)^{3/2}}, \cdots \right]$.

The recursive update law for $\bar{\eta}$ is then derived as:

$$\begin{aligned} \dot{\bar{\eta}} = P_\eta &\left(U(t)^\top B_0^\top y(t) - \omega_f \int_0^t \mathrm{e}^{-\omega_f(t-r)} U(r) B_0^\top y(r) \mathrm{d}r \right) \\ &+ \left(\omega_f P_\eta - P_\eta \left(\gamma \mathcal{G}(\bar{\eta}, \epsilon) + U(t) B_0^\top B_0 U(t) \right) P_\eta \right) \\ &\cdot \left(\int_0^t \mathrm{e}^{-\omega_f(t-r)} U(r) B_0^\top y(r) \mathrm{d}r \right) \end{aligned} \tag{14}$$

$$\dot{\bar{\eta}} = P_\eta U(t)^\top B_0^\top (y - B_0 U(t) \bar{\eta}) - \gamma P_\eta \mathcal{G}(\bar{\eta}, \epsilon) \cdot \bar{\eta} \tag{15}$$

which integrated over t yields the minimum solution to (10).

2) Stability Analysis: The control allocation matrix A is designed such that $B_0[\hat{\eta}]A = \mathbb{I}$. The closed loop dynamics are

$$\begin{aligned} \dot{x} &= f(x) + B_0 u + B_0([\eta] - I)u + r(x,u,t) + d(t) \\ &= -K\tilde{x} + \dot{x}_d - B_0[\tilde{\eta}]u + r(x,u,t) + d(t) \end{aligned} \tag{16}$$

$$\dot{\tilde{x}} = -K\tilde{x} - B_0 U \tilde{\eta} + r(x,u,t) + d(t) \tag{17}$$

where $\tilde{\eta} = \hat{\eta} - \eta$, $[\tilde{\eta}] = \operatorname{diag}(\tilde{\eta})$, and $\tilde{x} = x - x_d$.

By direct online optimization (e.g., the solution given in (12)), the ℓ_1 sparsity condition is fully enforced for m control inputs. Numerical integration of (13) and (15) is subject to integration errors and violation of the sparsity condition, though eradicates the need for online optimization.

Theorem 1. *For bounded disturbance d, the tracking error $\tilde{x}$ using u_{FT}, given by (7) and (15), will be exponentially bounded in the sense of finite-gain $\mathcal{L}_p$ stability [37].*

Proof. Consider the following Lyapunov function: $\mathcal{V} = \begin{bmatrix} \tilde{x} & \tilde{\eta} \end{bmatrix}^\top \mathcal{M} \begin{bmatrix} \tilde{x} & \tilde{\eta} \end{bmatrix}$ where $\mathcal{M} = \begin{bmatrix} I & 0 \\ 0 & P_\eta^{-1} \end{bmatrix}$. The derivative is computed as follows:

$$\begin{aligned} \dot{\mathcal{V}} &= 2\tilde{x}^\top \dot{\tilde{x}} + 2\tilde{\eta}^\top P_\eta^{-1} \dot{\tilde{\eta}} + \tilde{\eta}^\top \frac{\mathrm{d}\left(P_\eta^{-1}\right)}{\mathrm{d}t} \tilde{\eta} \\ &= - \begin{bmatrix} \tilde{x} \\ \tilde{\eta} \end{bmatrix}^\top \begin{bmatrix} 2K & B_0 U \\ UB_0^\top & U^\top B_0^\top B_0 U + \gamma \mathcal{G}(\bar{\eta}, \epsilon) + \omega_f P_\eta^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x} \\ \tilde{\eta} \end{bmatrix} \\ &\quad + 2 \begin{bmatrix} \tilde{x} \\ \tilde{\eta} \end{bmatrix}^\top \begin{bmatrix} r + d \\ U^\top B_0^\top (r + \delta) - \gamma \mathcal{G}(\bar{\eta}, \epsilon) \cdot \eta - P_\eta^{-1} \dot{\eta} \end{bmatrix} \end{aligned} \tag{18}$$

Since P_η^{-1} being bounded and uniformly positive definite, there exists some $\alpha > 0$ such that

$$-\begin{bmatrix} 2K & B_0 U \\ UB_0^\top & U^\top B_0^\top B_0 U + \gamma \mathcal{G}(\bar{\eta}, \epsilon) + \omega_f P_\eta^{-1} \end{bmatrix} \leq -2\alpha \mathcal{M} \tag{19}$$

Note that (19) is satisfied if $2K - 2\alpha I > 0$ and

$$UB_0^\top B_0 U + \gamma \mathcal{G} + (\omega_f - 2\alpha) P_\eta^{-1} - UB_0^\top \left(2K - 2\alpha I\right)^{-1} B_0 U > 0 \tag{20}$$

A sufficient condition for (20) is $2K - 2\alpha I - I > 0$, $\omega_f - 2\alpha \geq 0$ since $P_\eta^{-1} > 0$ is true by definition, $\gamma \mathcal{G}(\bar{\eta}, \epsilon) > 0$ and $UB_0^\top B_0 U - UB_0^\top \left(2K - 2\alpha I\right)^{-1} B_0 U > 0$ are also true. Using (18), (19) and the Cauchy-Schwartz inequality, $\dot{\mathcal{V}}$ can be bounded as follows:

$$\dot{\mathcal{V}} \leq -2\alpha \mathcal{V} + 2\sqrt{\mathcal{V}} D, \tag{21}$$

where $D = \left\| \begin{bmatrix} r + d \\ P_\eta^{1/2} UB_0^\top (r + \delta) - \gamma P_\eta^{1/2} \mathcal{G}(\bar{\eta}, \epsilon) \eta - P_\eta^{-1/2} \dot{\eta} \end{bmatrix} \right\|$.

Consider the related system $\mathcal{W}$ where $\mathcal{W} = \sqrt{\mathcal{V}}$ and $2\mathcal{W}\dot{\mathcal{W}} = \dot{V}$. Then, from the Comparison Lemma [37]

$$\sqrt{\mathcal{V}} \leq w(t) \leq \mathrm{e}^{-\alpha t} \left(w(0) - \sup_t \frac{D(t)}{\alpha} \right) + \sup_t \frac{D(t)}{\alpha} \tag{22}$$

Hence, $\sqrt{\mathcal{V}}$ and $\|\tilde{x}\|$ exponentially converge to the ball $\sup_t \frac{D}{\alpha}$. □

Remark 1. *All that remains is to show that $\sup_t D$ exists. By assumption, d, η, r, δ, and $\dot{\eta}$ are uniformly bounded and B_0, γ, ω_f, and ϵ are constants. Thus, D is uniformly bounded if $P_\eta^{1/2}$, $P_\eta^{-1/2}$, $\bar{\eta}$, and U are initially bounded and continuous. $P_\eta^{1/2}$ and $P_\eta^{-1/2}$ are uniformly bounded if P_η^{-1} is uniformly positive definite and uniformly bounded, respectively, which, in turn is true by the positiveness of ϵ and by the uniform boundedness of U and $\bar{\eta}$, respectively. U is uniformly bounded if $\tilde{x}$ is uniformly bounded and $(B_0\hat{\eta})^{\text{-}R}$ is uniformly bounded.*

Thus, the system is stable if $(B_0[\hat{\eta}])^{\text{-}R}$ exists for $\hat{\eta}$. Since $\hat{\eta}$ minimizes (8), (8) is a continuous function of the regularization parameter, γ, and $\hat{\eta} \to 1$ as $\gamma \to \infty$, there exists some γ for which $(B_0[\hat{\eta}])^{\text{-}R}$ is uniformly bounded. For small γ, $P_\eta^{1/2} U B_0^\top (r+\delta)$ dominates D. Thus, there is an inherent design trade off between the degree of regularization and the nominal modeling errors not captured by the effectiveness adaptation model. In fact, this trade off requires the inclusion of the learned dynamics discussed in Sec. III-B–III-C.

3) Discrete Update Law: The analogous discrete time least squares cost function for time step k is

$$\begin{aligned} J_k(\bar{\eta}) &= \sum_{i=0}^{k} e^{-\omega_f(t_k-t_i)} \left(y_i - \hat{g}\right)^\top \left(y_i - \hat{g}\right) + \gamma \left\|\bar{\eta}\right\|_1 \qquad (23) \\ &= \sum_{i=0}^{k} e^{-\omega_f(t_k-t_i)} \left(y_i^\top y_i - 2y_i^\top \hat{g} + \hat{g}^\top \hat{g}\right) + \gamma \left\|\bar{\eta}\right\|_1 \\ &= \left(\sum_{i=0}^{k} e^{-\omega_f(t_k-t_i)} y_i^\top y_i\right) - 2\left(\sum_{i=0}^{k} e^{-\omega_f(t_k-t_i)} y_i^\top B_0 U_i\right)\bar{\eta} \\ &\quad + \bar{\eta}^\top \left(\sum_{i=0}^{k} e^{-\omega_f(t_k-t_i)} U_i B_0^\top B_0 U_i\right)\bar{\eta} + \gamma \left\|\bar{\eta}\right\|_1 \end{aligned}$$

where $\hat{g} = \hat{g}_{\text{FT}}$ from (7). (23) is a convex function of $\bar{\eta} = \hat{\eta} - 1$. Therefore, the optimal $\hat{\eta}$ can be easily solved for using a number of optimization problem-solving tools such as [38], [39], enforcing a hard constraint to ensure $\hat{\eta} \in [0,1]^m$. During online computation, new measurements are incorporated by scaling the old summation terms by the exponential forgetting term $e^{(-\omega_f(t_k - t_{k-1}))}$ and adding the k-th term. Thresholding and linear smoothing is applied to the optimally solved effectiveness factors such that

$$\hat{\eta}_i = \begin{cases} 1 & \text{if } \hat{\eta}_i \geq \zeta \\ \frac{\hat{\eta}_i - \xi}{\zeta - \xi} & \text{if } \xi \leq \hat{\eta}_i < \zeta \\ 0 & \text{if } \hat{\eta}_i < \xi \end{cases} \qquad (24)$$

where ζ and ξ are upper and lower threshold parameters, respectively, such that $\zeta, \xi \in [0,1]$ and $\zeta > \xi$. This allows for a more aggressive update of $B_0[\hat{\eta}](t)$ by treating actuators with a sufficiently low effectiveness estimate as completely failed, and treating those with a sufficiently high effectiveness as fully functional, with smooth shaping inbetween for stability. Additionally, this output shaping captures the behavior of real-world motor failures as motors are highly unlikely to be functional but significantly degraded.

B. Neural-Fly Dynamics Adaptation for Real-Time Learning

The performance of effectiveness ($\bar{\eta}$) adaptation based on the discrete time cost function (23) degrades when the aircraft is in forward flight as the aerodynamic forces introduced increase model mismatch. To address this, NFFT utilizes [1], combining sparse failure identification with the learned model residuals and online adaptation of Neural-Fly to form (6).

Consider Neural-Fly's learned residual model of the time-varying aerodynamic forces, and its adaptive controller

$$\hat{g}_{\text{NF}}(x,u,t) = \phi(x,u)\hat{a}(t), \qquad (25)$$
$$u = u_{\text{NF}} = B_0^{\text{-}R}(-K\tilde{x} + \dot{x}_d - f(x) - \phi\hat{a}). \qquad (26)$$

For both u_{NFFT} (6) and u_{NF} (26), the basis function $\phi(x,u)$ is trained through the DAIML algorithm [1] during the off-line learning phase, while the time-varying linear weights $\hat{a}$ are updated online using the Kalman filter-based composite adaptation law and covariance update equation

$$\dot{\hat{a}} = -\lambda\hat{a} - P_a\phi^\top R^{-1}(\phi\hat{a} - y) + P_a\phi^\top\tilde{x} \qquad (27)$$
$$\dot{P}_a = -2\lambda P_a + Q - P_a\phi^\top R^{-1}\phi P_a \qquad (28)$$

where P_a is a covariance-like matrix for automatic gain tuning, and λ, R, and Q can be tuned similar to Kalman filtering.

Although $\hat{g}_{\text{NF}}$ alone yields reasonable aerodynamic residual force estimates and position tracking performance during nominal flight, actuator failures cause the plant to be severely altered, resulting in erroneous force estimates and flight performance. [36] showed that spectral normalization of a learned residual model for g guarantees the existence of a stabilizing control solution and robustness of the augmented control with [1] extending the learned-residual control framework to online learning of dynamics, allowing effective and robust adaptation of a pretrained model to a time-varying conditions. The learned model of the dynamics is incorporated into the control scheme via the iteratively updated control law for iteration step k

$$\begin{aligned} u_k &= B_0^{\text{-}R}\tau_{d,k}, \\ \tau_{d,k} &= \dot{x}_d - K\tilde{x} - f(x) - \hat{g}_{\text{NF}}(x, u_{k-1}, t) \end{aligned} \qquad (29)$$

where the fixed point iteration $\hat{g}_{\text{NF}}(x,u_{k-1},t) \approx \hat{g}_{\text{NF}}(x,u_k,t)$ is used to capture the non-affine control problem that arises from the learned model. To ensure stability, the fixed point iteration requires that the Lipschitz constant of the iteration be less than one, true when

$$\sigma(B_0^{\text{-}R}) \cdot \mathcal{L}_u(\hat{g}(x,u,t)) < 1 \qquad (30)$$

where $\sigma(B_0^{\text{-}R})$ is the spectral norm of $B_0^{\text{-}R}$ and $\mathcal{L}_u(\hat{g}(x,u,t))$ is the Lipschitz constant of $\hat{g}(x,u,t)$ with respect to u. Furthermore, [36] showed that the exponential convergence rate of the closed-loop system depends on the convergence rate of this fixed-point iteration. In particular, the exponential convergence rate, α, is proportional to $(\lambda_{\max}(K) - \rho)$, where ρ bounds the one-step difference in the control input, such that $\|u_k - u_{k-1}\| \leq \rho\|\tilde{x}_k\| \approx \rho\|\tilde{x}_{k-1}\|$. For actuator failures, the Lipschitz constant is given by $\mathcal{L}_u(g(x,u,t)) = \max_{i,t}(\eta_i - 1)$, therefore, the effective learned model of the fault dynamics will have a large Lipschitz constant. Because of this Lipschitz constant requirement, the control law in (29) can break down.

C. NFFT (u_{NFFT}) Combined from u_{FT} and u_{NF}

The main NFFT method, given in (5) and (6), combines (7), which estimates the actuator effectiveness and dynamically updates the corresponding control allocation, and (26), which estimates the aerodynamic residual forces through offline learning and composite adaptation. The discrete-time least-squares cost function (23) is still valid for this combined approach, except now $\hat{g} = \hat{g}_{\mathrm{NFFT}}$ yielding (5). This approach allows the force model to be updated in real-time, strengthening (23) by accounting for previously unmodeled aerodynamics disturbance forces.

Theorem 2. *Similar to Theorem 1 and (18), the tracking error $\tilde{x}$ using u_{NFFT} of (6) will be exponentially bounded.*

Proof. The derivative of the Lyapunov function $\mathcal{V}$ for u_{NFFT} is now extended to take the form

$$\begin{aligned}\dot{\mathcal{V}} &= 2\tilde{x}^\top\dot{\tilde{x}} + 2\tilde{\eta}^\top P_\eta^{-1}\dot{\tilde{\eta}} + 2\tilde{a}^\top P_a^{-1}\dot{\tilde{a}} + \tilde{\eta}^\top \dot{P}_\eta^{-1}\tilde{\eta} + \tilde{a}^\top \dot{P}_a^{-1}\tilde{a} \\ &= -\begin{bmatrix}\tilde{x}\\ \tilde{\eta}\\ \tilde{a}\end{bmatrix}^\top \begin{bmatrix} 2K & 0 & 0 \\ 0 & U^\top B_0^\top B_0 U + \gamma\mathcal{G} + \omega_f P_\eta^{-1} & \phi^\top (I+R^{-1})B_0 U \\ 0 & U^\top B_0^\top (I+R^{-1})\phi & \phi^\top R^{-1}\phi + P_a^{-1} Q P_a^{-1}\end{bmatrix}\begin{bmatrix}\tilde{x}\\ \tilde{\eta}\\ \tilde{a}\end{bmatrix} \\ &\quad + 2\begin{bmatrix}\tilde{x}\\ \tilde{\eta}\\ \tilde{a}\end{bmatrix}^\top \begin{bmatrix} d \\ U^\top B_0^\top \delta - \gamma\mathcal{G}\cdot\eta - P_\eta^{-1}\dot{\eta} \\ \phi^\top R^{-1}\delta - P_a^{-1}\lambda a - P_a^{-1}\dot{a}\end{bmatrix}\end{aligned} \tag{31}$$

where $\tilde{x} = x - x_d$, $\tilde{\eta} = \hat{\eta} - \eta$, and $\tilde{a} = \hat{a} - a$. The bounded input and bounded output stability (finite-gain $\mathcal{L}_p$ stability with sufficiently large Q) follows similar to the stability proof of u_{FT}. This follows from $\phi^\top R^{-1}\phi + P_a^{-1}QP_a^{-1} - 2\alpha P_a^{-1} > \frac{1}{\kappa} U B_0^\top (I+R^{-1})\phi\phi^\top (I+R^{-1})B_0 U$ and κ is the lower-bound of the first 2×2 given in (19) and since Q, λ can be chosen such that $P_a^{-1}QP_a^{-1} - 2\alpha P_a^{-1} > 0$, since $P_a^{-1} \geq 2\lambda/\|Q\| \cdot I$ if $P_a(0) \leq \frac{Q}{2\lambda}$ by Theorem 3. □

Theorem 3. $P_a^{-1} \geq 2\lambda/\|Q\| \ \cdot I$, $\forall t$ *if* $P_a(0) \leq \frac{Q}{2\lambda}$.

Proof. Consider $\mathcal{V}_{P_a}(t,z) = z^\top P_a z$, for some z s.t. $\|z\| \equiv 1$.
If $P_a(0) \leq \frac{Q}{2\lambda}$, then,

$$\begin{aligned}\dot{\mathcal{V}}_{P_a} &= z^\top(-2\lambda P_a + Q - P_a\phi^\top R^{-1}\phi P_a)z \\ &\leq z^\top(-\lambda P_a + Q)z \implies \mathcal{V}_{P_a} \leq \|\tfrac{Q}{2\lambda}\|\|z\|^2 = \|\tfrac{Q}{2\lambda}\|.\end{aligned}$$

Therefore, $P_a^{-1} \geq 2\lambda/\|Q\| \cdot I$ as required. □

D. Control Allocation through Online Optimization

From (3), $B_0{}^{\text{-}R}$ acts as a control allocation matrix that maps the required forces and torques to the control input u, and for over-actuated systems, is not unique. For a given actuation matrix, B, such as $B_0[\hat{\eta}]$, the control allocation matrix A must be found such that ideally $A = B^{\text{-}R}$ and $BA = I$, where both A and B may be time-varying, in order to prevent control-induced cross-coupling between the control axes while achieving the expected control efforts.

A natural choice for the control allocation matrix is the Moore-Penrose right pseudoinverse, which yields the minimum norm control input given any desired torque command. However, the pseudoinverse does not account for actual power usage or control saturation, leading to potentially non-actionable solutions, especially for asymmetric systems (such as when a motor fails).

For a symmetric configuration, the allocation matrix A that yields the maximum available control effort along each control axis independently is,

$$A_{\text{mca}} = \text{sign}(B^\top)\left(B\text{sign}(B^\top)\right)^{-1}. \tag{32}$$

The first term selects the correct sign to achieve the desired control affect, and the inverse term normalizes the response so that $BA = I$. Under a single motor failure, where $B = B_0[\eta](t)$, $(B_0[\eta])\text{sign}((B_0[\eta])^\top)$ will become non-diagonal, leading to cross-coupling in the different control axis and substantially degraded tracking performance.

Lastly, the allocation algorithm proposed in [22] imposes constraints on the cost function to ensure that the solution is a valid control allocation matrix for B with non-negative thrust factors. However, this method may yield some thrust factors equal to 0 with non-zero torque factors under certain motor failure configurations, leading to infinitesimally small torque commands resulting in control saturation.

Due to the limitations of prior approaches, a new allocation algorithm is proposed that directly maximizes the available torques (and hence control authority) in each axis while taking into account actuator saturation and minimizing the cross-coupling due to asymmetry in the actuator geometry. Further, by maximizing the control authority available, the chance of vehicle recovery during the failure is improved.

The control allocation problem is posed as a convex linear program, allowing it to be solved efficiently in real time with a numerical solver. The control signals for m actuators $u \in [0,1]^m$ are determined via $u = A_{\mathrm{FT}}\tau_d$ where τ_d is the desired force and torques and the actual force and torques generated, τ, is given by $\tau = Bu$. Let A_T, A_R, A_P, A_Y be the columns of A that produce required actuator command given desired vertical thrust, roll torque, pitch torque, and yaw torque, and let B_T, B_R, B_P, B_Y be the corresponding rows of B that govern the vertical thrust, and roll, pitch, and yaw torques. To generate roll-only torque without generating a net force, $u = hA_T \pm A_R$ where h is the 'hover throttle' required to offset the weight of the vehicle. Multiplying this by B_R gives the roll torque generated by the vehicle, and by applying the same method for the remaining axes, then summing the results,

$$\begin{aligned}\bar{A}_{\mathrm{FT}} = &\operatorname*{argmax}_A \sum_{i\in\{R,P,Y\}} B_i(hA_T + A_i) - \sum_{i\in\{R,P,Y\}} B_i(hA_T - A_i) \\ &+ B_T(hA_T) - \|A - A_0\|_F \\ \text{s.t.}\quad & \left.\begin{aligned} B_{R,P,Y}A_T = 0,\ 0 \leq A_T \leq 1,\end{aligned}\right\}\ \text{thrust constraint} \\ & \left.\begin{aligned} B_{T,P,Y}(hA_T + A_R) &= [h,0,0]^\top, \\ 0 \leq (hA_T \pm A_R) &\leq \mathbb{1},\end{aligned}\right\}\ \text{roll constraint} \\ & \left.\begin{aligned} B_{T,R,Y}(hA_T + A_P) &= [h,0,0]^\top, \\ 0 \leq (hA_T \pm A_P) &\leq \mathbb{1},\end{aligned}\right\}\ \text{pitch constraint} \\ & \left.\begin{aligned} B_{T,R,P}(hA_T + A_Y) &= [h,0,0]^\top, \\ 0 \leq (hA_T \pm A_Y) &\leq \mathbb{1},\end{aligned}\right\}\ \text{yaw constraint}\end{aligned} \tag{33}$$

The 'hover throttle' is applied as the $B_T(hA_T)$ term and the regularization term $\|A - A_0\|_F$ uses the Frobenius norm to stabilize the solution by penalizing deviations from a pre-computed nominal allocation matrix A_0 determined from (for example) (32). Finally, cross coupling and actuation limits are

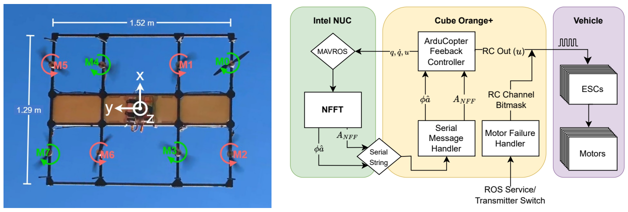

Fig. 2. Bottom view of the test aircraft with motor numbering (left) and the software schematic showing how the allocation and feedforward terms are sent to the flight controller (right).

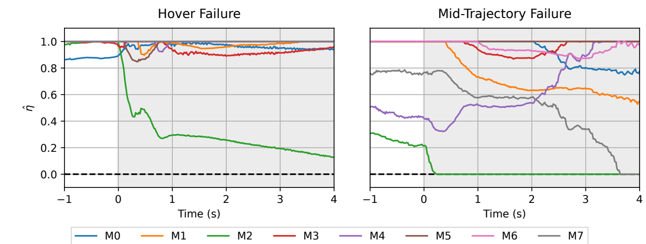

Fig. 3. Motor effectiveness estimate $\hat{\eta}$ without learned residual aerodynamic forces following rear left outer (M2) failure during hover (left) and during forward flight (right). The shaded areas show the failure region.

applied as hard constraints. For failure scenarios, $B\bar{A}_{\text{FT}} \neq I$, although $B\bar{A}_{\text{FT}}$ is diagonal. Thus, similar to (32), $\bar{A}_{\text{FT}}$ is scaled to get the final control allocation matrix A_{FT} where

$$A_{\text{FT}} = \bar{A}_{\text{FT}} \left(B\bar{A}_{\text{FT}}\right)^{-1}, \tag{34}$$

satisfying the condition $BA_{\text{FT}} = I$. Under nominal conditions, this exactly reproduces the solution from (32). As the algorithm computes A_{FT} based on the estimate of B, it will maintain the maximum control authority while preserving the axis-independent performance characteristics of the system for any degree of actuator degradation across any number of actuators, so long as the system remains controllable.

IV. Experimental Validation

A. Hardware Setup

NFFT was flown on a custom-built aircraft (Fig. 2, left) designed to emulate the 8-lifter-rotor configuration of the Autonomous Flying Ambulance UAV [40] (Fig. 1). The aircraft has a mass of $9.4\,\text{kg}$, a thrust-to-weight ratio of approximately 3.3, and is stabilized using a Cube Orange+ running ArduCopter 4.4.

NFFT is run onboard on a NUC (AMD Ryzen 7 4800U CPU, 32 GB RAM) with ROS 2 and MAVROS used to interface to the Cube's measurement data (Fig. 2, right). NFFT transfers the control allocation matrix and learned residual forces through a dedicated UART link which interfaces with ArduCopter using Lua script hooks. Lua scripting is also used to trigger motor failures either from ROS or the R/C transmitter. The only change required to the base ArduCopter 4.4 code was to allow setting the PID feedforward terms in the position controllers via Lua scripting.

The adaptation loop operates on a $50\,\text{Hz}$ timer, while the control allocation matrix updates at $10\,\text{Hz}$. With the experimental hardware setup, NFFT runs consistent loop times with the NFFT adaptation loop incorporating the latest state measurements in $0.38\,\text{ms}$ ($\sigma = 5.5\,\mu\text{s}$), computing the control actuation matrix and residual forces in $8.4\,\text{ms}$ ($\sigma = 43\,\mu\text{s}$), and solving of the control allocation matrix in $5.6\,\text{ms}$ ($\sigma = 52\,\mu\text{s}$). The convex functions are solved using Disciplined Parametrized Programming (DPP) in CVXPY [38], enhancing the efficiency of repeated computations.

B. Experimental Performance

The representation of the unmodeled aerodynamic forces, ϕ, was trained using DAIML [1] with data collected during a 10-minute, manually flown flight along an arbitrary trajectory with the baseline ArduCopter controller. Velocity, quaternion attitude, and control commands were recorded at $50\,\text{Hz}$ and were used as the input states. Importantly, the training dataset did not include any motor failures.

The experiments covered two scenarios: a single motor failure during hover and a single motor failure while tracking a steady level flight trajectory at $7.5\,\text{m/s}$ ground speed in low winds. Additionally, 'do no harm' tests were performed to show that under nominal operating conditions, NFFT did not degrade attitude or position tracking when compared to the baseline. In total, four control methods were evaluated: the baseline controller with ArduCopter's classical PID response; the Neural-Fly controller (26) with online composite aerodynamic residual force adaptation; the NFFT controller integrating Neural-Fly, adaptive failure identification, and control allocation through online optimization; and finally, the omniscient controller, which uses perfectly sensed failures and the NFFT's control allocation method. The pseudoinverse allocation method was not tested experimentally as simulations showed the solution oscillates erratically during the failure transient, causing potentially catastrophic behaviors in the aircraft response. Each failure test was repeated a minimum of 3 times for each controller for each motor to demonstrate repeatability, with similar results seen between each of the motor quadrants. Examples of the conducted experiments are shown in the supplemental video.

Control performance was evaluated by analyzing the aircraft attitude deviation from trim following each failure. Unless otherwise stated, all figures show failures of the rear left outer motor (M2) as it is the most severe failure tested, however similar responses are observed across all eight motor failure scenarios with inner motor failures typically seeing lower amplitude responses than outer motor failures.

Although the discrete ℓ_1-regularized actuator effectiveness adaptation (23) with a simple model of the unknown dynamics (7) is able to estimate the actuator effectiveness during hover failure (Fig. 3, left), increased model mismatch is observed when the aircraft begins to generate larger aerodynamic forces, severely deteriorating the actuator effectiveness estimates (Fig. 3, right). During these forward-flight tests, the actuator effectiveness estimates were so poor that the tests often needed to be aborted to recover the aircraft (Fig. 4).

The addition of the learned disturbance residuals enables NFFT to correctly identify the failed motor during trajectory tracking, estimating a lowered effectiveness exclusively for the failed motor (Fig. 5(a)). With this knowledge of the plant change, NFFT quickly reallocates actuator contributions

Fig. 4. Onboard footage shows the extreme departure from the desired attitude during outboard failures when NFFT is not running.

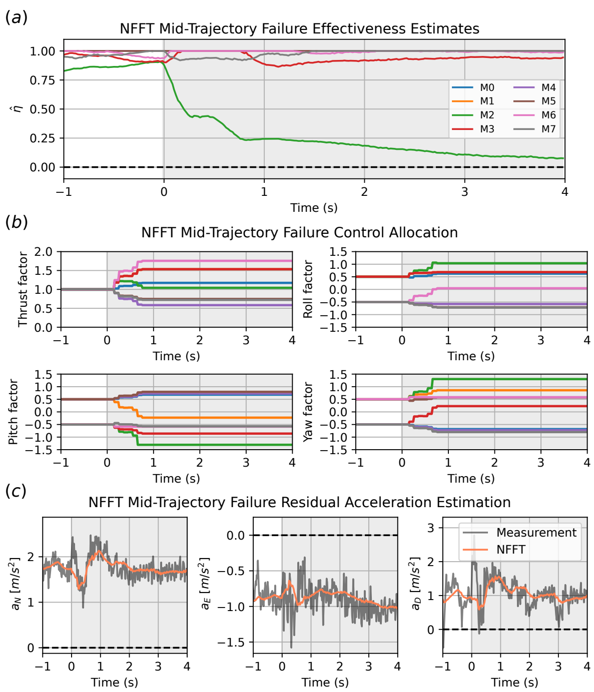

Fig. 5. Full NFFT adaptation and control allocation performance during motor failure while tracking a trajectory at ground speed $7.5\,\mathrm{m/s}$. The shaded areas show the failure region. (a) Motor effectiveness $\hat{\eta}$ estimate, (b) computed control allocation factors, (c) $\phi\hat{a}$ estimate compared to the measured forces.

among the remaining motors to maximize control authority and keep the axes decoupled, solving steady state allocation factors in approximately $0.66\,\mathrm{s}$ (Fig. 5(b)). The measured forces, as computed by the difference between the expected forces from the nominal model and the recorded inertial measurements, closely match the online aerodynamic residual force estimates $\phi\hat{a}$ (Fig. 5(c)), indicating very little model mismatch, even during the failure transient, hence the improved failure identification.

The aircraft attitude response following inboard and outboard actuator failures during forward flight (Fig. 6, Table I) shows NFFT behaves consistently between trials and notably reduces the time taken to bring the angle error back to near-zero. The initial peak offset of the angle offset remains high as NFFT requires deviation from nominal to detect the failure. With the perfect state knowledge used by the omniscient controller, the transient is all but eliminated as the control allocation matrix (33) is instantaneously recomputed to match the new actuator configuration. Finally, the Neural-Fly controller, which relies only on residual force adaptation, exhibits degraded attitude tracking performance compared to the baseline as the failure drastically alters the plant model.

TABLE I
MEAN PEAK ATTITUDE ERRORS DURING FAILURES

Controller	Peak R Error mean	Peak R Error stdev	Peak P Error mean	Peak P Error stdev	Peak Y Error mean	Peak Y Error stdev
Inboard						
Baseline	4.38°	0.33°	7.79°	0.25°	5.97°	1.31°
Neural-Fly	4.37°	0.45°	7.24°	0.41°	6.63°	0.89°
NFFT	3.98°	0.11°	6.43°	0.28°	5.24°	0.19°
Omniscient	2.01°	0.29°	2.74°	0.27°	1.69°	0.27°
Outboard						
Baseline	8.37°	0.11°	8.13°	0.95°	12.88°	1.84°
Neural-Fly	9.92°	0.28°	9.51°	0.90°	15.69°	3.27°
NFFT	7.45°	0.55°	7.38°	1.04°	6.99°	0.62°
Omniscient	3.57°	0.58°	3.36°	0.90°	1.91°	0.37°

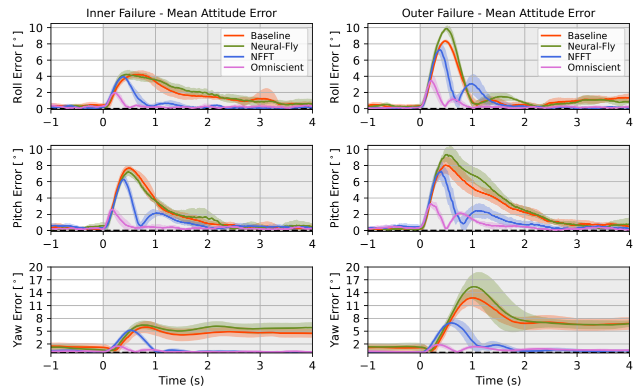

Fig. 6. Baseline, Neural-Fly, NFFT, and failure-omniscient controller attitude tracking performance for M3 (inner) and M2 (outer) motors. Solid curves show the mean tracking error over four 5-second failure events during forward flight, with standard deviations shaded in corresponding colors. The gray shaded areas show the failure region.

The learned residual force $\phi\hat{a}$ with feedforward cancellation leads to improvement in aircraft position tracking performance under actuator failure only when combined with the failure detection (Fig. 7(a), NFFT). If the motor failure is not accounted for, the abrupt change in the system plant significantly degrades the controller's ability to track the desired position (Fig. 7(a), Neural-Fly), demonstrating the need for fault-tolerant control schemes that take into account plant changes when using NN-based residual adaptation methods such as Neural-Fly. Neural-Fly, which was effective for wind-based residuals [1], causes a $74\,\%$ degradation in mean position tracking error compared to the baseline, while the combined NFFT approach yields a $48\,\%$ improvement, similar to that of the omniscient controller (Fig. 7(b)). The NFFT position tracking performance is on par with that of the omniscient controller due to fast settling time of NFFT, demonstrating the effectiveness of NFFT at mitigating the motor failures despite no direct sensing of the failure.

V. Conclusion

This paper presents a new method for detecting and compensating for motor failures in multirotor aircraft without direct sensing. This method, coupled with a deep-learning based adaptive controller, Neural-Fly, is able to simultaneously predict unmodeled forces and motor failures, enabling precise tracking even under plant changes and model uncertainties. The proposed sparse failure identification method successfully

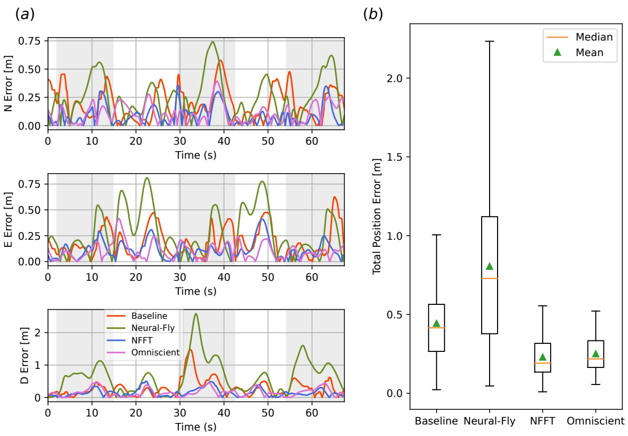

Fig. 7. Position error during rectangular trajectory tracking with outer motor failures. (a) Measured position errors while flying at $7.5\,\mathrm{m/s}$. Failures are represented by gray regions and are toggled at three different points during the experiment. (b) Data distribution from the corresponding flights.

isolates a failed motor within one second and reallocates actuator effort without any direct observation of the failure, while running in conjunction with adaptive flight control for tracking a desired trajectory. The maximum-control-authority allocation scheme showed improvements over prior methods in maximizing control authority amidst faults and potential actuator saturation. This end-to-end FTC method represents a large progress in developing robust fault detection and compensation strategies for aircraft, enabling ever-safer skies.

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